

Real-Time Helmet and Triple Riding Violation Detection Using YOLOv11s and ByteTrack for Smart Traffic Surveillance

Abstract

Road traffic accidents caused by helmet violations and triple riding remain a significant concern, leading to severe injuries and loss of life. Traditional traffic monitoring methods rely heavily on manual observation, making them inefficient for continuous surveillance. This paper proposes an AI-based smart traffic surveillance system for the real-time detection of helmet violations and triple riding using YOLOv11s and ByteTrack. YOLOv11s is employed to accurately detect motorcycles, riders, helmets, and license plates, while ByteTrack is used to maintain consistent tracking of vehicles across consecutive video frames. The system analyzes the number of riders on each motorcycle and identifies riders without helmets. Upon detecting a violation, the license plate is recognized using Optical Character Recognition (OCR), and the violation details, including the vehicle number, date, time, and evidence image, are stored in a centralized database. A digital violation report is generated to support traffic enforcement, with future integration of an automated e-Challan system. The proposed approach aims to improve road safety, reduce manual monitoring efforts, and provide an efficient solution for intelligent traffic surveillance.

Keywords: YOLOv11s, ByteTrack, Helmet Detection, Triple Riding Detection, Smart Traffic Surveillance, OCR, License Plate Recognition.

1. Introduction

Road traffic accidents are one of the leading causes of fatalities worldwide, with a significant number of incidents occurring due to riders not wearing helmets and violations such as triple riding. According to traffic safety regulations, wearing a helmet and limiting the number of riders on a motorcycle are essential for reducing the risk of severe injuries. However, monitoring these violations manually through traffic personnel is time-consuming, labor-intensive, and prone to human error, especially in densely populated urban areas.

Recent advancements in computer vision and deep learning have enabled intelligent traffic surveillance systems capable of detecting road traffic violations automatically. Object detection models such as the YOLO (You Only Look Once) family have demonstrated high accuracy and real-time performance in detecting vehicles, riders, helmets, and other road objects. Similarly, multi-object tracking algorithms such as ByteTrack provide reliable tracking of moving vehicles across video frames, enabling continuous monitoring without duplicate detections.

Although several existing studies have proposed helmet detection systems, many focus only on identifying helmet violations. Some systems detect passengers but do not accurately identify triple riding, while others lack vehicle tracking or automated violation reporting. In addition, many existing approaches do not integrate license plate recognition and violation management into a single framework.

To address these limitations, this paper proposes a Real-Time Helmet and Triple Riding Violation Detection System using YOLOv11s and ByteTrack. The proposed system detects motorcycles, riders, helmets, and license plates from live surveillance video. ByteTrack is used to assign unique identities to motorcycles and track them across video frames. The system analyzes helmet usage and counts the number of riders to identify traffic violations. Once a violation is detected, the license plate is recognized using OCR, and the violation details are stored in a database for report generation. This integrated approach enhances the efficiency of traffic monitoring while reducing the dependency on manual supervision.

The major contributions of this work are:

- Development of a real-time helmet and triple riding detection system using YOLOv11s.
- Integration of ByteTrack for accurate multi-object tracking.
- Detection and recognition of vehicle license plates using OCR.
- Automatic generation of digital violation reports for traffic authorities.
- A scalable framework that supports future integration with automated e-Challan systems.